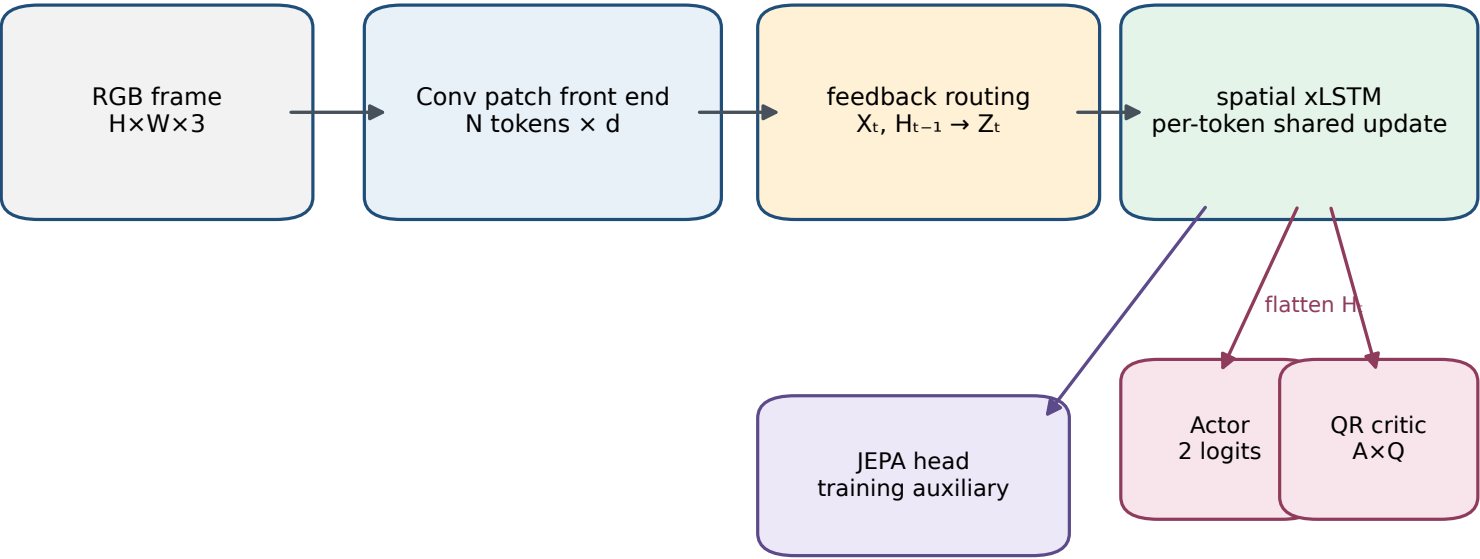


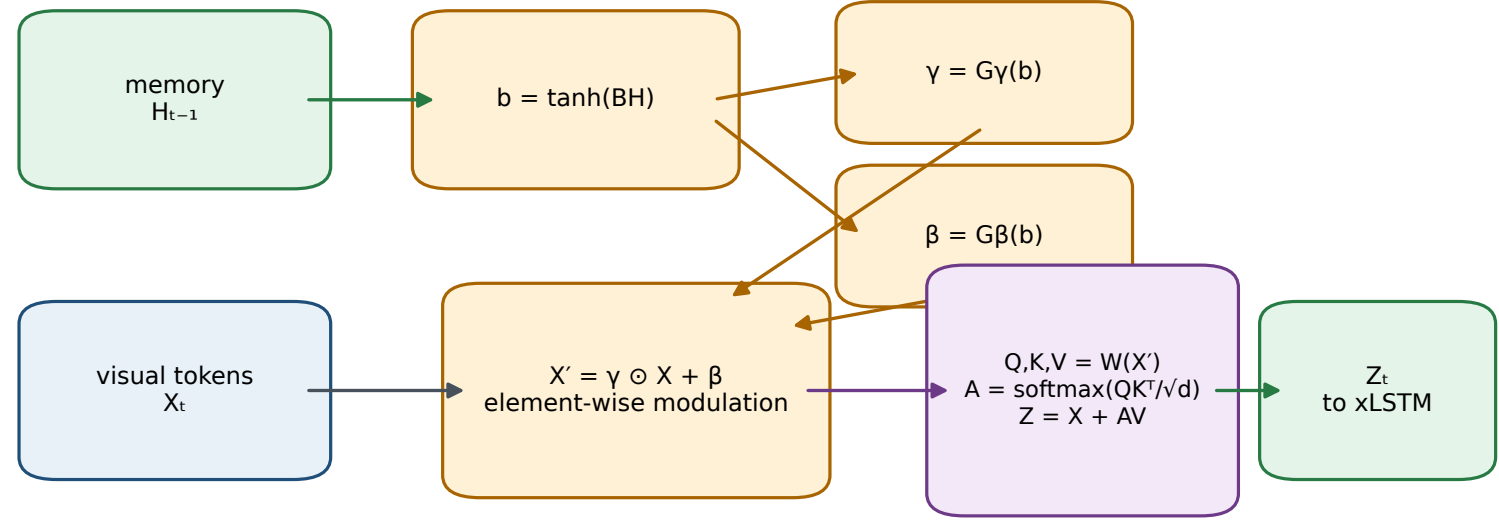
M2 · Recurrent VDA architecture and admitted feedback families

A Shared recurrent VDA agent



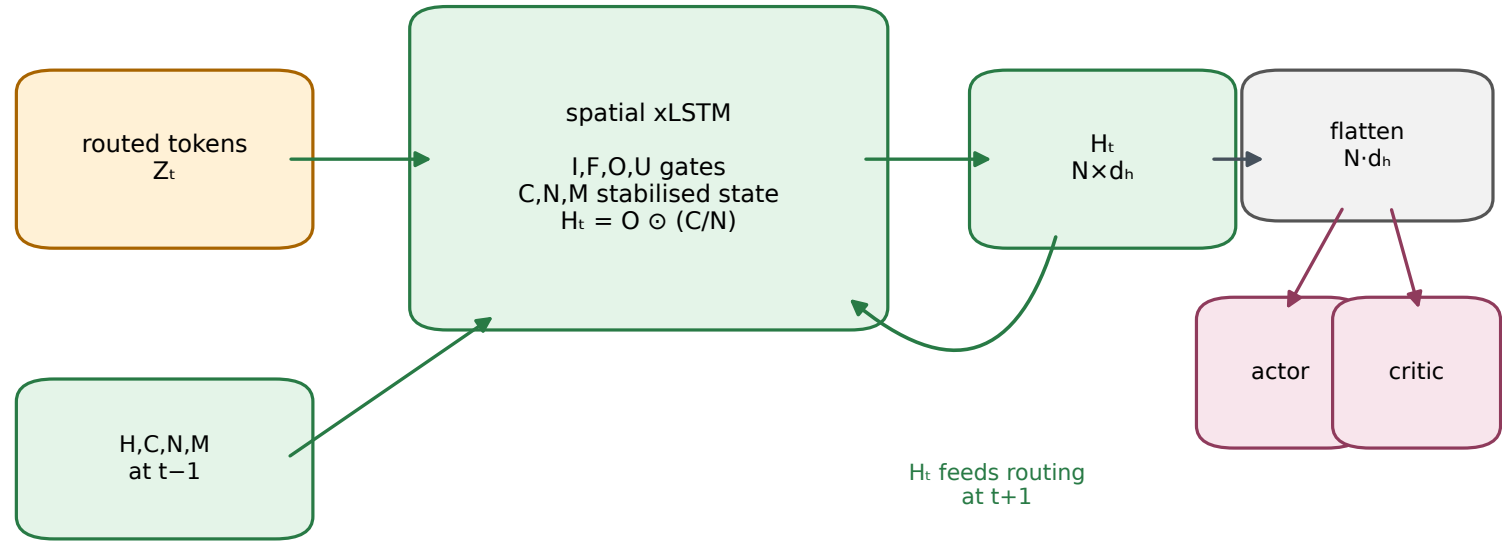
N follows the task geometry (2, 4, 9, or 16); d and memory width are run-resolved.

B Element-wise affine feedback (affine_ew)



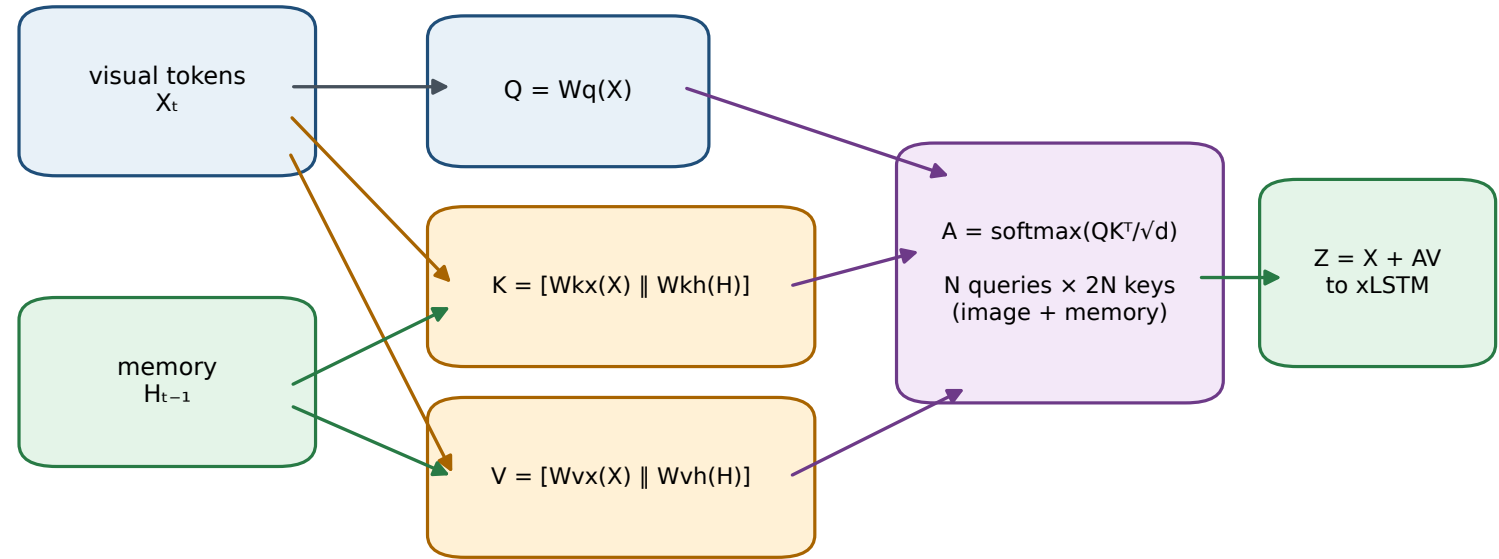
Initialization: $\gamma = 1$ and $\beta = 0$, so routing starts as plain self-attention.

C Spatial recurrent memory and readouts



Patch structure is preserved through routing and memory, then removed at the RL readout.

D Cross-attention comparator (crossattn1)



The same recurrent memory and actor/critic heads follow; only the routing operator changes.